

# Programming for Biomedical Informatics

## Week 2 - FTP Bulk Download & APIs

This week we're going to practice using bulk download and API in a practical example. This week we are going to be looking at a resource called GenCC - The Gene Curation Collection - <https://thegencc.org>

GenCC is primarily focussed on providing a unified source of gene-disease relationships that is harmonised across the various international efforts that operate in this space. In the process of doing so they are establishing standards for curation and annotation to maximise the consistency and useability of their data.

Our efforts today will focus on a full FTP download of the GenCC data, some parsing, and use of the API at Bioportal to gather more information about some of the elements of the GenCC data. This is a **slightly** contrived example, but hopefully it is both interesting and useful to give you an idea of how these two approaches can be used together.

```
In [2]: # Step One - Bulk Download of GenCC data

'''The url to the latest version of the data is here - https://search.thege

#pseudocode 1 - download the data using the urllib libraryd
'''###YOUR CODE HERE###'''
import urllib.request

url = "https://search.thegencc.org/download/action/submissions-export-tsv"
urllib.request.urlretrieve(url, "gencc_data.tsv")

#pseudocode 2 - read the data into a pandas dataframe *you can see from the
'''###YOUR CODE HERE###'''
import pandas as pd
df = pd.read_csv("gencc_data.tsv", sep="\t")

#pseudocode 3 - look ay the first few rows of the dataframe
'''###YOUR CODE HERE###'''
df.head()
```

Out [2]:

|   | uuid                                                          | gene_curie | gene_symbol | disease_curie | disease_title                                              |
|---|---------------------------------------------------------------|------------|-------------|---------------|------------------------------------------------------------|
| 0 | GENCC_000101-<br>HGNC_10896-<br>OMIM_182212-<br>HP_0000006... | HGNC:10896 | SKI         | MONDO:0008426 | Shprintzen-<br>Goldberg syndrome                           |
| 1 | GENCC_000101-<br>HGNC_16636-<br>OMIM_171300-<br>HP_0000006... | HGNC:16636 | KIF1B       | MONDO:0008233 | pheochromocytoma                                           |
| 2 | GENCC_000101-<br>HGNC_16636-<br>OMIM_118210-<br>HP_0000006... | HGNC:16636 | KIF1B       | MONDO:0007308 | Charcot-Marie-<br>Tooth disease type<br>2A1                |
| 3 | GENCC_000101-<br>HGNC_17939-<br>OMIM_617532-<br>HP_0000007... | HGNC:17939 | SLC45A1     | MONDO:0044322 | intellectual<br>developmental<br>disorder with<br>neuro... |
| 4 | GENCC_000101-<br>HGNC_11071-<br>OMIM_616291-<br>HP_0000007... | HGNC:11071 | SLC9A1      | MONDO:0014572 | Lichtenstein-Knorr<br>syndrome                             |

5 rows × 30 columns

In [14]:

```
# Step Two – Gene Counts by Disease

'''Now we want a summary of how many genes have been annotated to each disease'''

#pseudocode 4 – group the data by disease and count the number of genes for each disease
'''###YOUR CODE HERE###'''
grouped = df.groupby('disease_title').agg({'gene_symbol': 'nunique'}).reset_index()

#pseudocode 5 – sort the data in descending order
'''###YOUR CODE HERE###'''
grouped = grouped.sort_values(by='gene_symbol', ascending=False)

#pseudocode 6 – look at the first few rows of the dataframe
'''###YOUR CODE HERE###'''
grouped.head()

'''There's a nice package called PrettyTable that can be used to print the contents of a dataframe'''

from prettytable import PrettyTable

#make some simple two-column data
data = [['disease1', 1000], ['disease2', 2000], ['disease3', 3000], ['disease4', 4000]]

# create a table
table = PrettyTable()

# add the columns
table.field_names = ['Disease Name', 'Number of Genes']
```

```

# add the rows
for i in range(len(data)):
    table.add_row(data[i])

# print the table
print(table)

# pseudocode 7 – create a table using the PrettyTable package
'''###YOUR CODE HERE###'''
table = PrettyTable()
table.field_names = ["Disease Name", "Number of Genes"]
for index, row in grouped.iterrows():
    table.add_row([row['disease_title'], row['gene_symbol']])
print(table[:10]) # print only the first 10 rows for brevity

'''There is a very nice plotting package called seaborn that can be used to
Here's an example of how to create a horizontal bar chart using seaborn'''

#plot a horizontal bar chart of the data
import seaborn as sns
import pandas as pd

# set the style
sns.set_style('darkgrid')

# plot the data
sns.barplot(x='Number of Genes', y='Disease Name', data=pd.DataFrame(data, c

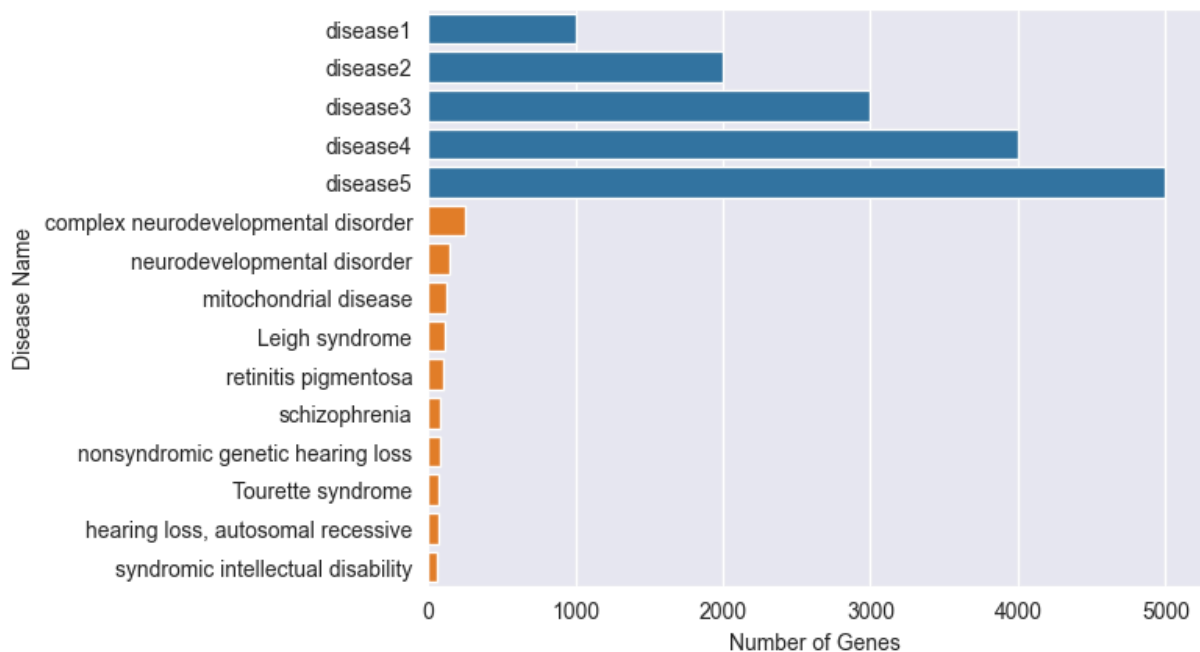
#pseudocode 8 – plot the 10 diseases with the most genes
'''###YOUR CODE HERE###'''
top10 = grouped.head(10)
sns.barplot(x='gene_symbol', y='disease_title', data=top10)

```

| Disease Name | Number of Genes |
|--------------|-----------------|
| disease1     | 1000            |
| disease2     | 2000            |
| disease3     | 3000            |
| disease4     | 4000            |
| disease5     | 5000            |

| Disease Name                        | Number of Genes |
|-------------------------------------|-----------------|
| complex neurodevelopmental disorder | 255             |
| neurodevelopmental disorder         | 145             |
| mitochondrial disease               | 127             |
| Leigh syndrome                      | 115             |
| retinitis pigmentosa                | 102             |
| schizophrenia                       | 88              |
| nonsyndromic genetic hearing loss   | 86              |
| Tourette syndrome                   | 76              |
| hearing loss, autosomal recessive   | 73              |
| syndromic intellectual disability   | 68              |

Out[14]: <Axes: xlabel='Number of Genes', ylabel='Disease Name'>



In [8]: *# Step Three – Gene–Disease Confidence*

```
'''When making decisions about gene disease relationships there are a comple
of assertions. These are stored in the 'classification_title' field of the or

#pseudocode 9 – group the data by classification_title (confidence categorie
'''###YOUR CODE HERE###'''
grouped = df.groupby('classification_title').agg({'gene_symbol': 'nunique'})

#pseudocode 10 – sort the data in descending order
```

```

'''###YOUR CODE HERE###'''
grouped = grouped.sort_values(by='gene_symbol', ascending=False)

#pseudocode 11 - look at the first few rows of the dataframe
'''###YOUR CODE HERE###'''
grouped.head()

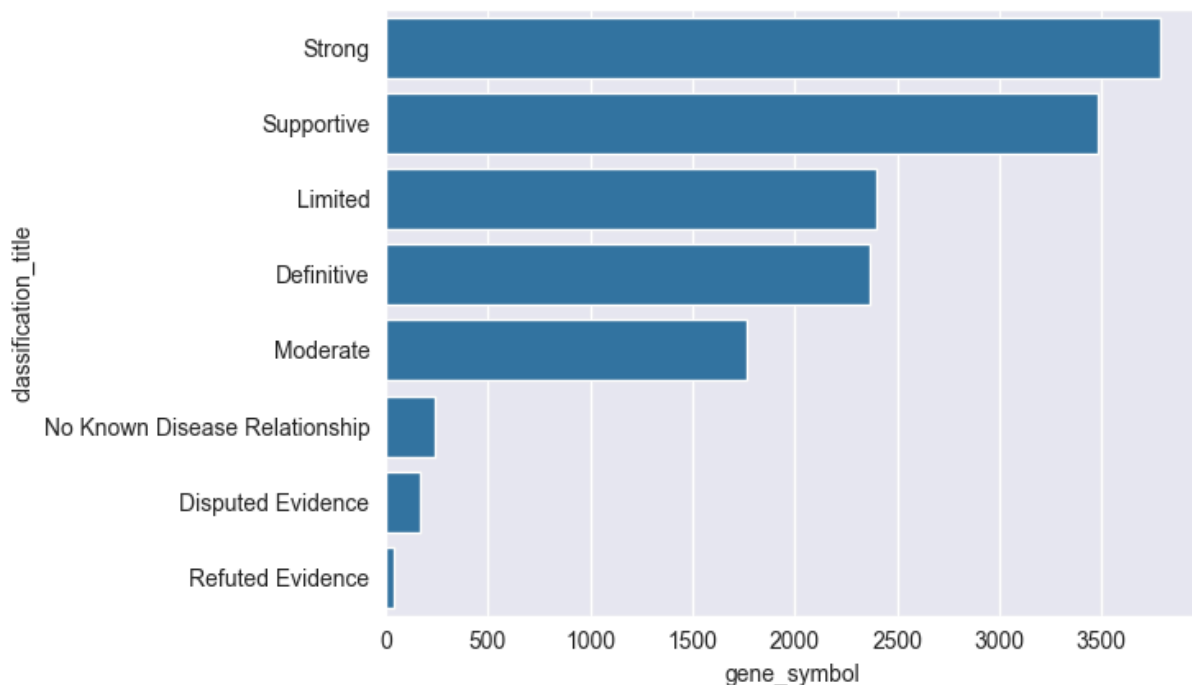
#pseudocode 12 - create a table using the PrettyTable package
'''###YOUR CODE HERE###'''
table = PrettyTable()
table.field_names = ["Classification Title", "Number of Genes"]
for index, row in grouped.iterrows():
    table.add_row([row['classification_title'], row['gene_symbol']])
print(table)

#pseudocode 13 - plot the classification_titles with the most genes
'''###YOUR CODE HERE###'''
top10 = grouped.head(10)
sns.barplot(x='gene_symbol', y='classification_title', data=top10)

```

| Classification Title          | Number of Genes |
|-------------------------------|-----------------|
| Strong                        | 3790            |
| Supportive                    | 3479            |
| Limited                       | 2399            |
| Definitive                    | 2363            |
| Moderate                      | 1765            |
| No Known Disease Relationship | 239             |
| Disputed Evidence             | 172             |
| Refuted Evidence              | 36              |

Out[8]: <Axes: xlabel='gene\_symbol', ylabel='classification\_title'>



In [11]: *# Step Four – Modes of Inheritance*

```
'''Many of these diseases are Mendelian diseases. This means that they are c
GenCC captures this information in the 'moi_curie' field'''

#pseudocode 14 – group the data by moi_curie (modes of inheritance) and cour
'''###YOUR CODE HERE###'''
grouped = df.groupby('moi_curie').agg({'gene_symbol': 'nunique'}).reset_inde
grouped = grouped.rename(columns={'moi_curie': 'Mode of Inheritance Curie',
print(grouped)

'''This is good but the values we have here are accession ids that aren't ve
We know that terms (classes) like 'HP:0000006' are actually from the Human P
We're going to use the BioPortal API to get infromation about these classes
```

|   | Mode of Inheritance Curie | Number of Genes |
|---|---------------------------|-----------------|
| 0 | HP:0000005                | 879             |
| 1 | HP:0000006                | 2772            |
| 2 | HP:0000007                | 3398            |
| 3 | HP:0001417                | 277             |
| 4 | HP:0001419                | 2               |
| 5 | HP:0001427                | 43              |
| 6 | HP:0001450                | 3               |
| 7 | HP:0032113                | 118             |

Out[11]: "This is good but the values we have here are accession ids that aren't ver  
y informative\nWe know that terms (classes) like 'HP:0000006' are actually  
from the Human Phenotype Ontology (HPO)\nWe're going to use the BioPortal  
API to get infromation about these classes and then finish by merging these  
data with our table"

In [12]: '''first we need an API key from BioPortal. You can create an account on Bio  
<https://bioportal.bioontology.org/>  
For this example I will use my API key – note how I have it stored in a file

```
#load my API key for BioPortal
with open("../api_keys/bioportal.txt", "r") as file:
    apikey = file.read().strip()

'''Here is an example of how you can use the requests library to get informa

#find details for an ontology term using the BioPortal API
import requests
import json

#set the URL for the BioPortal API – here we query with a fixed HP class id
url = 'http://data.bioontology.org/ontologies/HP/classes/HP:0000006?apikey='

#send the request
response = requests.get(url)

#parse the JSON in the response
class_info = response.json()

#display the JSON (for interest!)
print(json.dumps(class_info, indent=2))
```

```
#display the keys in the JSON (for interest!)
print(class_info.keys())

#display the values for the keys
print(class_info['@id'],class_info['prefLabel'])
print(class_info['definition'][0])
```

```

{
  "prefLabel": "Autosomal dominant inheritance",
  "synonym": [
    "Autosomal dominant form",
    "Autosomal dominant type",
    "Autosomal dominant",
    "monoallelic_autosomal"
  ],
  "definition": [
    "A mode of inheritance that is observed for traits related to a gene encoded on one of the autosomes (i.e., the human chromosomes 1-22) in which a trait manifests in heterozygotes. In the context of medical genetics, an autosomal dominant disorder is caused when a single copy of the mutant allele is present. Males and females are affected equally, and can both transmit the disorder with a risk of 50% for each child of inheriting the mutant allele."
  ],
  "cui": [],
  "semanticType": [],
  "obsolete": false,
  "created": null,
  "modified": null,
  "memberOf": [],
  "inScheme": [],
  "@id": "http://purl.obolibrary.org/obo/HP_0000006",
  "@type": "http://www.w3.org/2002/07/owl#Class",
  "links": {
    "self": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006",
    "ontology": "https://data.bioontology.org/ontologies/HP",
    "children": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/children",
    "parents": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/parents",
    "descendants": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/descendants",
    "ancestors": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/ancestors",
    "instances": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/instances",
    "tree": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/tree",
    "notes": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/notes",
    "mappings": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/mappings",
    "ui": "http://bioportal.bioontology.org/ontologies/HP?p=classes&conceptid=http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006",
    "@context": {
      "self": "http://www.w3.org/2002/07/owl#Class",
      "ontology": "http://data.bioontology.org/metadata/Ontology",
      "children": "http://www.w3.org/2002/07/owl#Class",
      "parents": "http://www.w3.org/2002/07/owl#Class",
      "descendants": "http://www.w3.org/2002/07/owl#Class",
      "ancestors": "http://www.w3.org/2002/07/owl#Class",
      "instances": "http://data.bioontology.org/metadata/Instance",
      "tree": "http://www.w3.org/2002/07/owl#Class",

```



```

        "notes": "http://data.bioontology.org/metadata/Note",
        "mappings": "http://data.bioontology.org/metadata/Mapping",
        "ui": "http://www.w3.org/2002/07/owl#Class"
    }
},
"@context": {
    "@vocab": "http://data.bioontology.org/metadata/",
    "prefLabel": "http://www.w3.org/2004/02/skos/core#prefLabel",
    "synonym": "http://www.w3.org/2004/02/skos/core#altLabel",
    "definition": "http://www.w3.org/2004/02/skos/core#definition",
    "obsolete": "http://www.w3.org/2002/07/owl#deprecated",
    "semanticType": "http://bioportal.bioontology.org/ontologies/umls/hasST
Y",
    "cui": "http://bioportal.bioontology.org/ontologies/umls/cui",
    "inScheme": "http://www.w3.org/2004/02/skos/core#inScheme",
    "created": "http://purl.org/dc/terms/created",
    "modified": "http://purl.org/dc/terms/modified",
    "@language": "en"
}
}
dict_keys(['prefLabel', 'synonym', 'definition', 'cui', 'semanticType', 'obsolete', 'created', 'modified', 'memberOf', 'inScheme', '@id', '@type', 'links', '@context'])
http://purl.obolibrary.org/obo/HP_000006 Autosomal dominant inheritance
A mode of inheritance that is observed for traits related to a gene encoded on one of the autosomes (i.e., the human chromosomes 1-22) in which a trait manifests in heterozygotes. In the context of medical genetics, an autosomal dominant disorder is caused when a single copy of the mutant allele is present. Males and females are affected equally, and can both transmit the disorder with a risk of 50% for each child of inheriting the mutant allele.

```

```

In [13]: #pseudocode 15 - using the unique accession ids in the moi_curie field, get
'''###YOUR CODE HERE###'''
moi_classes = {}
for moi in grouped['Mode of Inheritance Curie'].unique():
    url = f'http://data.bioontology.org/ontologies/HP/classes/{moi}?apikey={
    response = requests.get(url)
    if response.status_code == 200:
        class_info = response.json()
        moi_classes[moi] = {
            'prefLabel': class_info.get('prefLabel', 'N/A'),
            'definition': class_info.get('definition', ['N/A'])[0]
        }
    else:
        moi_classes[moi] = {'prefLabel': 'N/A', 'definition': 'N/A'}

#pseudocode 16 - merge the data with the moi_curie field with the data from
# the mode of inheritance and the number of genes for each and the name of t
'''###YOUR CODE HERE###'''
grouped['Mode of Inheritance Name'] = grouped['Mode of Inheritance Curie'].n
print(grouped)

```

|   | Mode of Inheritance | Curie      | Number of Genes | Mode of Inheritance Name        |
|---|---------------------|------------|-----------------|---------------------------------|
| e |                     |            |                 |                                 |
| 0 |                     | HP:0000005 | 879             | Mode of inheritance             |
| e |                     |            |                 |                                 |
| 1 |                     | HP:0000006 | 2772            | Autosomal dominant inheritance  |
| e |                     |            |                 |                                 |
| 2 |                     | HP:0000007 | 3398            | Autosomal recessive inheritance |
| e |                     |            |                 |                                 |
| 3 |                     | HP:0001417 | 277             | X-linked inheritance            |
| e |                     |            |                 |                                 |
| 4 |                     | HP:0001419 | 2               | X-linked recessive inheritance  |
| e |                     |            |                 |                                 |
| 5 |                     | HP:0001427 | 43              | Mitochondrial inheritance       |
| e |                     |            |                 |                                 |
| 6 |                     | HP:0001450 | 3               | Y-linked inheritance            |
| e |                     |            |                 |                                 |
| 7 |                     | HP:0032113 | 118             | Semidominant inheritance        |
| e |                     |            |                 |                                 |

Well done!

You've now practised FTP download, API use and also completed some data manipulation using Pandas, PrettyTable, and plotting with Seaborn